

# THE STACKING NUMBER OF A TREE

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**ABSTRACT.** The stacking number of a graph is the least integer  $t \geq 2$  such that every configuration of  $t$  pebbles can be transformed by pebbling moves into a configuration supported on one vertex. We prove that, for every finite tree  $T$  with at least two vertices, this number equals the rooted distance-and-degree estimator conjectured by Csernák and Soukup. The proof uses an exact recursive characterization of stackability at a prescribed vertex, an explicit zero-score obstruction, and a weighted cancellation argument for arbitrary nonstackable configurations. The complete theorem is formalized in Lean.

## 1. INTRODUCTION

A pebbling move removes two pebbles from a vertex and puts one pebble on an adjacent vertex. Csernák and Soukup [2] introduced the stacking number: how many pebbles guarantee that all remaining pebbles can be brought to a single vertex, whose location may depend on the configuration? They conjectured an explicit formula for trees and verified it computationally for every tree of order at most seven.

For a finite tree  $T$  and  $r \in V(T)$ , put

$$\begin{aligned} (1) \quad \ell_T(r) &= |\{v \neq r : \deg_T(v) = 1\}|, \\ (2) \quad \sigma_T(r) &= 1 + \sum_{\substack{v \in V(T) \\ v=r \text{ or } \deg_T(v) > 1}} \deg_T(v) 2^{d_T(r,v)}, \\ (3) \quad E_T(r) &= \sigma_T(r) + \ell_T(r), \quad \text{estim}(T) = \max_{r \in V(T)} E_T(r). \end{aligned}$$

Here  $d_T$  denotes graph distance. We retain the source notation  $\sigma_T$  and  $\text{estim}$ ; our  $\ell_T(r)$  is its  $\text{leaf}(r)$ , and  $E_T$  abbreviates its rooted estimator. Conjecture 10.3 of [2] asserts  $\text{stack}(T) = \text{estim}(T)$ .

The definition of the stacking number minimizes over  $t \geq 2$ . Thus the one-vertex tree has stacking number 2, whereas the displayed estimator is 1. We state the conjecture with the explicit hypothesis  $|V(T)| \geq 2$ ; this isolates a boundary convention in the definitions.

**Theorem 1.1.** *For every finite tree  $T$  with at least two vertices,*

$$\text{stack}(T) = \text{estim}(T).$$

We prove the tree-stacking estimator conjecture of Csernák and Soukup for nontrivial trees. The proof has three structural ingredients. A branch sends an integer message across its boundary edge, recording the optimal gain or cost of clearing it. An empty branch is a separate state. These messages give an exact criterion  $S_r(C) > 0$  for stacking a configuration  $C$  at  $r$ , including necessity for arbitrary move order. An explicit configuration of mass  $\text{estim}(T) - 1$  has every score zero. For the reverse bound, the nonnegative defects  $-S_v(C)$  satisfy local two-sided edge equations. We assign the excess contributions to distinct endpoints and cancel them against weighted defects. A consolidation lemma then bounds the auxiliary height weights by distances from one root.

Classical graph pebbling prescribes a target vertex; see Chung [1] and Hurlbert [4]. Cover pebbling prescribes positive demands at all vertices. Sjöstrand's theorem [5] reduces

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its universal threshold to distance potentials of configurations concentrated at one vertex. In stacking, every vertex outside the eventual target must be empty. The messages below account for this exact clearing condition.

## 2. CONFIGURATIONS AND THE ESTIMATOR

All graphs in this paper are finite, simple and undirected. A tree is a nonempty connected acyclic graph. Unless expressly stated otherwise,  $T$  has at least two vertices. We write  $u \sim v$  for adjacency, and omit the subscript  $T$  from degrees and distances when no confusion is possible. The natural numbers include zero.

A *configuration* is a function  $C : V(T) \rightarrow \mathbb{N}$ ; its mass is  $|C| = \sum_v C(v)$  and its support is  $\text{supp}(C) = \{v : C(v) > 0\}$ . The source writes  $\|c\|$  for mass; we use  $|C|$ . The move  $u \rightarrow v$  is legal when  $u \sim v$  and  $C(u) \geq 2$ . It replaces  $C(u)$  by  $C(u) - 2$  and  $C(v)$  by  $C(v) + 1$ . Reachability permits a finite sequence of legal moves, including the empty sequence. Each move decreases mass by one.

A configuration is *stacked at  $r$*  when  $\text{supp}(C) = \{r\}$ . Write  $\text{StackableAt}(T, C, r)$  if a configuration stacked at  $r$  is reachable from  $C$ . The configuration is *stackable* if this holds for some  $r$ . In particular, the zero configuration is not stacked or stackable. The stacking number is the least integer  $t \geq 2$  for which every configuration of mass *exactly*  $t$  is stackable. Our proof will also establish the existence of such an integer for trees.

Let  $L$  be the number of degree-one vertices of  $T$ . The estimator has the following useful form, uniform in the root:

$$(4) \quad E_T(r) - 1 = L + \sum_{\deg(v) > 1} \deg(v) 2^{d(r,v)}.$$

If  $r$  is internal, this follows directly from the definition. If  $r$  is a leaf, its degree-one term in  $\sigma_T(r)$  replaces the leaf omitted from  $\ell_T(r)$ . Also

$$(5) \quad E_T(r) \geq |V(T)| + 1.$$

Indeed, in the expansion  $1 + \deg(r) + \ell_T(r) + \sum_{v \neq r, \deg(v) > 1} \deg(v) 2^{d(r,v)}$ , the root degree is at least one and each of the other  $|V(T)| - 1$  vertices contributes at least one.

## 3. BRANCH MESSAGES AND ROOTED STACKABILITY

For an oriented edge  $v \rightarrow p$ , let  $B_{v|p}$  be the component containing  $v$  after deleting  $vp$ . Its root is  $v$  and its external boundary vertex is  $p$ . The children are the branches  $B_{u|v}$  with  $u \sim v$ ,  $u \neq p$ . Call a branch *occupied* when its restriction of  $C$  is nonzero.

Define the integer transfer function by

$$(6) \quad F(x) = \begin{cases} 2x - 3, & x \leq 1, \\ x/2, & x \geq 2 \text{ even}, \\ (x - 3)/2, & x \geq 3 \text{ odd}. \end{cases}$$

In particular,  $F(2) = 1$  and  $F(3) = 0$ . Messages take values in the disjoint union  $\{\text{EMPTY}\} \sqcup \mathbb{Z}$ . For an empty branch put  $m_{v \rightarrow p} = \text{EMPTY}$ . For an occupied branch put

$$(7) \quad m_{v \rightarrow p} = d_{v \rightarrow p} := F \left( C(v) + \sum_{\substack{u \sim v, u \neq p \\ B_{u|v} \text{ occupied}}} d_{u \rightarrow v} \right).$$

The recursion terminates because each child branch has fewer vertices. Only integer messages are summed. When writing a sum over all neighbours we use the numerical

contribution  $\bar{d}_{u \rightarrow v}$ , equal to  $d_{u \rightarrow v}$  on occupied branches and 0 on empty branches. This convention for sums does not identify the states **EMPTY** and 0. The rooted score is

$$(8) \quad S_r(C) = C(r) + \sum_{u \sim r} \bar{d}_{u \rightarrow r}.$$

**Theorem 3.1** (Exact boundary invariant). *Let  $B = B_{v|p}$  be occupied with message  $d = d_{v \rightarrow p}$ . Work on the subtree induced by  $B \cup \{p\}$ , with initial configuration  $C \upharpoonright B$  on  $B$  and  $q \geq 0$  pebbles at  $p$ . A positive stack at  $p$  with  $B$  empty is reachable if and only if  $q + d > 0$ . In that case the greatest possible final pile at  $p$  is  $q + d$ .*

We prove necessity in a form that also applies when other branches interleave their moves at  $p$ .

**Lemma 3.2** (Boundary flux). *Suppose a legal sequence clears  $B_{v|p}$ . Let  $m_{ab}$  count its moves from  $a$  to  $b$ , and put  $\phi_{v|p} = m_{vp} - 2m_{pv}$ . If the branch is occupied initially, then*

$$\phi_{v|p} \leq d_{v \rightarrow p}, \quad \phi_{v|p} \equiv d_{v \rightarrow p} \pmod{3}.$$

*If it is empty initially, then  $\phi_{v|p} = -3k$  for some  $k \geq 0$ . Moves elsewhere in the tree are permitted.*

*Proof.* The move counts satisfy, at every cleared vertex  $w$ ,

$$(9) \quad C(w) + \sum_{z \sim w} m_{zw} - 2 \sum_{z \sim w} m_{wz} = 0.$$

For each initially occupied branch which is cleared, at least one move crosses its boundary outwards. Indeed, consider the move after which that branch becomes empty for the last time. An internal move leaves a pebble inside, and an inward move also does so. Thus this move is an outward boundary move. This observation applies to every occupied child branch, independently of the moves outside it.

First consider an initially empty branch. Sum (9) over its vertices. If  $I$  is the number of internal moves and  $a = m_{vp}$ ,  $b = m_{pv}$ , the result is  $b - 2a - I = 0$ . Hence its boundary flux is  $a - 2b = -3a - 2I \leq 0$ . It is divisible by three: give the boundary vertex sign +1 and alternate signs along tree edges. The signed sum of the balance equations over the branch is congruent to minus its boundary flux modulo three, because each internal move contributes a multiple of three. The initial and final signed sums on the empty branch vanish. This proves the assertion for empty branches, even when they are used temporarily.

For an occupied branch, induct on its size. By induction, each occupied child has flux  $d_i - 3k_i$  with  $k_i \geq 0$ , and each empty child has flux  $-3k_i$ . Put  $x = C(v) + \sum d_i$  and  $K = \sum k_i$ , where the first sum runs over occupied children. At  $v$ , equation (9) reads

$$y + b - 2a = 0, \quad y = x - 3K, \quad a = m_{vp} \geq 1, \quad b = m_{pv} \geq 0.$$

For any integer  $y$ , these conditions imply

$$(10) \quad a - 2b = 2y - 3a \leq F(y), \quad a - 2b \equiv F(y) \pmod{3}.$$

To see this, the least possible  $a$  is  $\max(1, \lceil y/2 \rceil)$ ; substituting this value gives precisely the three cases of (6). Any increase in  $a$  lowers the flux by three.

The transfer function satisfies

$$(11) \quad F(z - 3) \leq F(z), \quad F(z) \equiv 2z \pmod{3}.$$

For the inequality, the difference  $F(z) - F(z - 3)$  is 6 when  $z \leq 1$ , 6, 3, 3 at  $z = 2, 3, 4$ , respectively, and is 0 for odd  $z \geq 5$  and 3 for even  $z \geq 6$ . The congruence follows from each case in (6). Thus  $F(x - 3K) \leq F(x)$  and  $F(x - 3K) \equiv F(x) \pmod{3}$ . Apply (10) to finish the induction.  $\square$

*Proof of Theorem 3.1.* The final boundary pile in any allowed clearing sequence is  $q + \phi$ , so Lemma 3.2 proves necessity and the upper bound. For attainment we induct on the branch size.

We use the following scheduling observation. Suppose disjoint occupied child branches have gains  $d_i$ , and a vertex initially has  $a \geq 0$  pebbles. If  $a + \sum_i d_i > 0$ , process branches with positive gain first, then those with zero gain, then those with negative gain. A positive-gain task has positive final pile even when begun at zero. Before a zero-gain task the pile is positive, since the total remaining gains are nonpositive and the final total is positive. During the negative-gain tasks the pile after each task is at least the final positive total. The inductive boundary invariant therefore realizes every task and leaves exactly  $a + \sum_i d_i$ . Tasks touch only their branch and the common boundary; no other task's initial interior configuration is changed.

Let  $x = C(v) + \sum_i d_i$  for the occupied children of  $B$ . If  $x \geq 2$ , the observation clears the children and leaves  $x$  at  $v$ . For  $x = 2k$ , make  $k$  moves from  $v$  to  $p$ ; the final boundary pile is  $q + k = q + F(x)$ . For  $x = 2k + 1 \geq 3$ , make  $k$  moves from  $v$  to  $p$ , leaving one pebble at  $v$ . The assumption  $q + F(x) = q + k - 1 > 0$  implies  $q + k \geq 2$ . One move  $p \rightarrow v$  followed by one move  $v \rightarrow p$  is therefore legal; it clears  $v$  and leaves  $q + k - 1$  at  $p$ .

If  $x \leq 1$ , put  $j = 2 - x \geq 1$ . The assumption  $q + F(x) = q - 2j + 1 > 0$  implies  $q \geq 2j$ . Make  $j$  moves  $p \rightarrow v$ . The children's effective total at  $v$  is now  $x + j = 2$ . The observation clears them leaving two pebbles at  $v$ , and one final move  $v \rightarrow p$  gives pile  $q - 2j + 1 = q + F(x)$ . These constructions also cover the case with no children and complete the induction.  $\square$

**Theorem 3.3** (Exact rooted stackability). *For every configuration  $C$  on a finite tree and every vertex  $r$ ,*

$$\text{StackableAt}(T, C, r) \iff S_r(C) > 0.$$

*Consequently  $C$  is nonstackable if and only if all its scores are nonpositive.*

*Proof.* If the score is positive, apply the scheduling observation to the occupied branches incident with  $r$ , starting with  $C(r)$  pebbles. It gives a stack of size exactly  $S_r(C)$  at  $r$ . Conversely, in any sequence ending in a stack at  $r$ , every incident branch is cleared. Lemma 3.2 bounds the flux from each occupied branch by its message and from each empty branch by zero. The final pile is therefore at most  $S_r(C)$ , and it is positive. The final equivalence follows by quantifying over all targets.  $\square$

*Remark 3.4.* An occupied one-vertex branch containing three pebbles has message  $F(3) = 0$ . It can be cleared to its boundary only when the boundary has a positive initial pile. An empty branch may be left alone even when that pile is zero. This is why the recursive state must retain **EMPTY** separately from the integer zero.

#### 4. AN EXTREMAL ZERO-SCORE CONFIGURATION

Fix  $r \in V(T)$  and root the tree at  $r$ . Define

$$(12) \quad C_r(v) = \begin{cases} \sigma_T(r) - 1, & v = r, \\ 1, & v \neq r \text{ and } \deg(v) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

For a nonroot vertex  $v$ , write  $T_v$  for its descendant subtree and put

$$(13) \quad H_v = \begin{cases} 1, & v \text{ has no children,} \\ 3 + 2 \sum_{u \text{ child of } v} H_u, & \text{otherwise.} \end{cases}$$

These are positive integers, with closed form

$$(14) \quad H_v = 1 + 2 \sum_{\substack{w \in V(T_v) \\ \deg_T(w) > 1}} \deg_T(w) 2^{d(v,w)}.$$

For a leaf this is immediate. For a nonleaf  $v \neq r$ , the number of children is  $\deg_T(v) - 1$ . Substitution of the formula at its children gives

$$3 + 2 \sum_u H_u = 1 + 2 \deg_T(v) + 4 \sum_u \sum_{\substack{w \in V(T_u) \\ \deg_T(w) > 1}} \deg_T(w) 2^{d(u,w)},$$

which is (14) at  $v$ . Summing this closed form over the children of  $r$  yields

$$(15) \quad \sum_{u \text{ child of } r} H_u = \deg_T(r) + \sum_{\substack{w \neq r \\ \deg_T(w) > 1}} \deg_T(w) 2^{d(r,w)} = \sigma_T(r) - 1.$$

**Theorem 4.1.** *For the configuration  $C_r$ ,*

$$S_v(C_r) = 0 \quad (v \in V(T)), \quad |C_r| = E_T(r) - 1.$$

*In particular,  $C_r$  is nonstackable.*

*Proof.* Every descendant branch is occupied, since it contains a leaf other than  $r$ . The opposite side is occupied as well, since  $C_r(r) = \sigma_T(r) - 1 \geq \deg_T(r) \geq 1$ . Induction from the leaves gives upward messages  $d_{v \rightarrow p} = -H_v$ : at a leaf this is  $F(1) = -1$ ; at an internal nonroot vertex the effective input is  $-\sum_u H_u$ , so its message is  $-3 - 2\sum_u H_u = -H_v$ . Equation (15) implies  $S_r(C_r) = 0$ .

Now suppose  $p$  has score zero and  $v$  is its child. Removing the contribution  $-H_v$  from its score gives effective input  $H_v$  on the  $p$ -side of  $pv$ , hence  $d_{p \rightarrow v} = F(H_v)$ . If  $v$  is a leaf, its score is  $1 + F(1) = 0$ . Otherwise let  $K = \sum_{u \text{ child of } v} H_u \geq 1$ . Then  $H_v = 3 + 2K$ ,  $F(H_v) = K$ , and the score at  $v$  is  $-K + K = 0$ . Propagation away from  $r$  proves the assertion at every vertex. Finally, the total mass is  $\sigma_T(r) - 1 + \ell_T(r) = E_T(r) - 1$ . Nonstackability follows from Theorem 3.3.  $\square$

**Corollary 4.2.** *There is a nonstackable configuration of mass  $\text{estim}(T) - 1$  whose score is zero at every vertex.*

*Proof.* Choose a root maximizing  $E_T$  and apply Theorem 4.1.  $\square$

## 5. THE ARBITRARY-DEFECT UPPER BOUND

We now bound the mass of every configuration whose scores are all nonpositive. All degrees, paths and distances in this section belong to the original tree. In particular, the argument retains the empty branches: their contributions will be cancelled separately from the contributions of edges with occupied support on both sides.

Let  $T$  be a finite tree with at least two vertices, and let  $C$  satisfy  $S_v(C) \leq 0$  for every  $v$ . Set

$$\delta_v = -S_v(C) \geq 0.$$

If  $C$  is identically zero the desired bound is immediate. Henceforth choose an occupied vertex  $o$ , so that  $C(o) > 0$ . This vertex is used to assign edge owners; the estimator root obtained at the end may be different.

**5.1. Retained edges and the local defect equations.** For adjacent vertices  $u, v$ , write  $T_{u|v}$  for the component containing  $u$  after deleting  $uv$ . Call  $uv$  *retained* if both  $T_{u|v}$  and  $T_{v|u}$  contain an occupied vertex. Its two messages are then integers. For any edge, let

$$\bar{d}_{u \rightarrow v} = \begin{cases} d_{u \rightarrow v}, & T_{u|v} \text{ is occupied,} \\ 0, & T_{u|v} \text{ is empty.} \end{cases}$$

This notation records the numerical contribution of a message in a score; it does not identify the message **EMPTY** with the integer message zero. Thus

$$(16) \quad C(v) = -\delta_v - \sum_{u \sim v} \bar{d}_{u \rightarrow v}.$$

**Lemma 5.1** (Support separation). *The two sides of an edge partition  $V(T)$ . Every non-retained edge has exactly one empty side. A retained edge lies on a path between two occupied vertices, and every edge of a path between two occupied vertices is retained. More specifically, if  $uv$  is retained and  $x$  is an occupied vertex in  $T_{u|v}$ , every edge on the path from  $u$  to  $x$  is retained. If  $uv$  is retained and no other retained edge is incident with  $u$ , then  $C(u) > 0$ . Finally, an endpoint on the empty side of an edge is incident with no retained edge.*

*Proof.* Deleting an edge of a tree produces exactly two components, giving the partition. Since  $C(o) > 0$ , at least one component is occupied. If both are occupied, choose one occupied vertex in each; their unique path crosses the deleted edge. Conversely, deleting any edge on the path between two occupied vertices separates those vertices, so that edge is retained.

For the more specific assertion, choose an occupied vertex  $y$  in  $T_{v|u}$ . The path from  $x$  to  $y$  consists of the path from  $x$  to  $u$ , the edge  $uv$ , and the path from  $v$  to  $y$ . Every edge on its  $x$ - $u$  segment is therefore retained. If  $u$  is unoccupied, an occupied vertex  $x$  on its side is distinct from  $u$ , and the edge of that segment incident with  $u$  is a retained edge other than  $uv$ . This proves the next assertion by contraposition.

For the last assertion, let  $T_{v|u}$  be empty. The edge  $uv$  itself is nonretained. For every other neighbour  $w$  of  $v$ , the component  $T_{w|v}$  lies inside  $T_{v|u}$  and is empty. Hence  $wv$  is also nonretained.  $\square$

For a retained edge  $uv$ , put  $a = d_{u \rightarrow v}$  and  $b = d_{v \rightarrow u}$ . Removing the opposite message from each endpoint score gives the two effective branch inputs. The message recursion therefore yields

$$(17) \quad a = F(-\delta_u - b), \quad b = F(-\delta_v - a).$$

These equations are only asserted for retained edges.

**Lemma 5.2** (Classification of retained edge states). *The solutions of (17) with  $\delta_u, \delta_v \geq 0$  have exactly the following forms, up to exchanging  $u$  and  $v$ :*

(1) *An oriented state, with*

$$(18) \quad a = k \geq 0, \quad b = -2(k + \delta_v) - 3, \quad \delta_u = 2\delta_v \text{ or } \delta_u = 2\delta_v + 3.$$

*The latter alternative requires  $k > 0$ . We direct this edge  $u \rightarrow v$ .*

(2) *A two-negative state, with unique  $r, q \in \mathbb{N}$  such that*

$$(19) \quad a = -(2r + 1), \quad b = -(2q + 1), \quad \delta_u = r + 2q, \quad \delta_v = q + 2r.$$

*Both messages cannot be nonnegative.*

*Proof.* The definition of  $F$  gives

$$\begin{aligned} F(z) < 0 &\iff z \leq 1, \\ F(z) = -(2r+1) &\iff z = 1-r \quad (r \geq 0), \\ F(z) = k &\iff (k > 0 \text{ and } z = 2k) \text{ or } z = 2k+3 \quad (k \geq 0). \end{aligned}$$

Indeed, the first two statements use  $F(z) = 2z - 3$  for  $z \leq 1$ ; the even positive and odd positive cases of  $F$  give the third, including the unique preimage 3 of zero.

If  $a = k \geq 0$ , then  $-\delta_v - k \leq 0$ , so the second edge equation gives  $b = -2(k + \delta_v) - 3 < 0$ . Substituting this into the first equation and using the nonnegative preimages gives either  $-\delta_u + 2k + 2\delta_v + 3 = 2k + 3$ , or the same left side equals  $2k$  with  $k > 0$ . These are exactly the two defect alternatives in (18). The case  $b \geq 0$  is symmetric.

If both messages are negative, the first two identities express them uniquely as  $-(2r+1)$  and  $-(2q+1)$ , with  $r, q \geq 0$ . Their unique preimages give  $-\delta_u + 2q + 1 = 1 - r$  and  $-\delta_v + 2r + 1 = 1 - q$ , proving (19). Conversely, substitution verifies every displayed state, subject to the stated restriction when  $k = 0$ .  $\square$

Call an oriented state *defective* if its head defect  $\delta_v$  is positive. Call a two-negative state defective if  $r + q > 0$ . The remaining two-negative state is the neutral state  $(-1, -1)$ , with both endpoint defects zero. The defective retained edges form a forest, since they are a subset of the edges of  $T$ .

## 5.2. Owners, orientations and auxiliary weights.

**Lemma 5.3** (Injective incident owners). *For each edge  $e = uv$ , let  $\omega(e)$  be the endpoint farther from  $o$ . Then  $\omega$  is well defined and injective on  $E(T)$ . If  $T_{v|u}$  is empty, then  $\omega(uv) = v$ .*

*Proof.* The unique path from  $o$  crosses  $uv$  precisely when reaching the endpoint on the side not containing  $o$ . Thus the endpoint distances differ by one. The owner is the more distant endpoint. Each vertex other than  $o$  has exactly one neighbour preceding it on its unique path from  $o$ ; consequently it owns exactly that predecessor edge. In particular two edges cannot share an owner. If  $T_{v|u}$  is empty, the occupied vertex  $o$  lies on the  $u$  side, so  $v$  is the farther endpoint.  $\square$

Define an auxiliary partial orientation of  $T$  as follows. Keep the direction of every oriented state, including those with zero head defect. Direct each defective two-negative edge toward its owner. Leave neutral edges and nonretained edges unoriented. The classification ensures that no edge has both directions. Since  $T$  has no undirected cycle, this directed graph has no directed cycle.

Let  $h(v)$  be the maximum length of a directed path ending at  $v$ , allowing the path of length zero, and set

$$(20) \quad A_v = 2^{h(v)}.$$

The maximum exists because a directed path has no repeated vertex. For an auxiliary arrow  $u \rightarrow v$ , a longest path ending at  $u$  can be extended by this arrow: if it already contained  $v$ , it would give a directed cycle. Hence

$$(21) \quad h(v) \geq h(u) + 1, \quad A_v \geq 2A_u.$$

A vertex has positive height exactly when it has an incoming auxiliary arrow. Since every auxiliary arrow is retained, Lemma 5.1 also shows that the endpoint on the empty side of any edge has height zero and weight one.

**5.3. Local charges and global defect cancellation.** For an undirected edge  $e = uv$ , define its weighted message contribution by

$$E_e = -(A_v \bar{d}_{u \rightarrow v} + A_u \bar{d}_{v \rightarrow u}).$$

Define two nonnegative charges, with disjoint conditions of application:

$$Q_e^{\text{ret}} = \begin{cases} A_{\omega(e)} \delta_{\omega(e)}, & e \text{ is retained and defective,} \\ 0, & \text{otherwise,} \end{cases}$$

$$Q_e^{\text{emp}} = \begin{cases} A_{\omega(e)} \delta_{\omega(e)}, & e \text{ is nonretained,} \\ 0, & \text{otherwise.} \end{cases}$$

**Lemma 5.4** (Retained-edge charge). *For every retained edge  $uv$ ,*

$$E_{uv} \leq A_u + A_v + Q_{uv}^{\text{ret}}.$$

*Proof.* First consider an oriented state  $u \rightarrow v$  with parameter  $k \geq 0$ . By (18) and (21),

$$\begin{aligned} E_{uv} &= A_u(2(k + \delta_v) + 3) - A_v k \\ &= 3A_u + k(2A_u - A_v) + 2A_u \delta_v \\ &\leq A_u + A_v + 2A_u \delta_v. \end{aligned}$$

The excess  $2A_u \delta_v$  is zero when the edge is nondefective. Otherwise either endpoint can pay it: the local defect relation gives  $A_u \delta_u \geq 2A_u \delta_v$ , and doubling gives  $A_v \delta_v \geq 2A_u \delta_v$ . Thus it is bounded by the charge at the prescribed owner, regardless of whether that owner is the tail or head.

For a defective two-negative state, name its endpoints so that the auxiliary direction is  $u \rightarrow v$ , with  $v = \omega(uv)$ , and use the parameters in (19). Then

$$\begin{aligned} E_{uv} &= A_u + A_v + 2A_v r + 2A_u q \\ &\leq A_u + A_v + A_v(2r + q) \\ &= A_u + A_v + A_v \delta_v. \end{aligned}$$

Here  $q \geq 0$  and  $2A_u \leq A_v$  justify the inequality. A neutral edge has  $r = q = 0$  and contributes exactly  $A_u + A_v$ , completing the proof.  $\square$

**Lemma 5.5** (Empty-side cancellation). *If  $T_{v|u}$  is empty, then*

$$E_{uv} = A_v \delta_v = Q_{uv}^{\text{emp}}.$$

*Proof.* The message  $m_{v \rightarrow u}$  is literally EMPTY. At  $v$ , the initial pile is zero, and every incident branch other than the  $u$  side is empty. Consequently  $S_v(C) = d_{u \rightarrow v}$ , where this last message is an integer because the  $u$  side contains  $o$ . Thus  $\bar{d}_{v \rightarrow u} = 0$  and  $d_{u \rightarrow v} = -\delta_v$ , giving  $E_{uv} = A_v \delta_v$ . By Lemma 5.3,  $v$  is the owner. In addition  $A_v = 1$ , as observed above. No integer defect-state equations were applied to the empty message.  $\square$

**Proposition 5.6** (Weighted cancellation). *The auxiliary weighted mass satisfies*

$$(22) \quad \sum_v A_v C(v) \leq \sum_v \deg_T(v) A_v.$$

*Proof.* Grouping (16) by undirected edges gives the exact identity

$$(23) \quad \sum_v A_v C(v) = \sum_{e \in E(T)} E_e - \sum_v A_v \delta_v.$$

Lemmas 5.4 and 5.5 give, uniformly over all ambient edges,

$$E_{uv} \leq A_u + A_v + Q_{uv}^{\text{ret}} + Q_{uv}^{\text{emp}}.$$

Each sum of the two charges is at most its one owner budget  $A_{\omega(e)}\delta_{\omega(e)}$ . Owners are injective on all edges, and every vertex budget is nonnegative, so

$$\sum_e (Q_e^{\text{ret}} + Q_e^{\text{emp}}) \leq \sum_e A_{\omega(e)}\delta_{\omega(e)} \leq \sum_v A_v\delta_v.$$

Substitution in (23) cancels the budgets. Finally, each vertex weight occurs once for each incident edge, so  $\sum_{uv \in E(T)} (A_u + A_v) = \sum_v \deg_T(v)A_v$ .  $\square$

**5.4. Removing leaf weights.** Let  $L$  be the number of ambient vertices of degree one.

**Lemma 5.7** (Ambient leaf slack). *Every degree-one vertex  $v$  with  $A_v > 1$  is occupied. Consequently*

$$(24) \quad |C| \leq L + \sum_{\deg_T(v) > 1} \deg_T(v)2^{h(v)}.$$

*Proof.* If  $A_v > 1$ , then  $h(v) > 0$ , so there is an incoming auxiliary arrow at  $v$ . Its underlying edge is retained. As  $v$  has degree one, its side of that edge is the singleton  $\{v\}$ ; retention implies  $C(v) \geq 1$ . For any degree-one vertex, therefore,

$$A_v - 1 \leq (A_v - 1)C(v) :$$

if  $A_v = 1$  both sides are zero, and otherwise  $C(v) \geq 1$  proves it. Since  $A_v \geq 1$  at every vertex, all configuration slack terms are nonnegative. Using Proposition 5.6,

$$\begin{aligned} |C| &= \sum_v A_v C(v) - \sum_v (A_v - 1)C(v) \\ &\leq \sum_v \deg_T(v)A_v - \sum_{\deg_T(v)=1} (A_v - 1) \\ &= L + \sum_{\deg_T(v) > 1} \deg_T(v)A_v. \end{aligned}$$

The last equality uses that every vertex of a nontrivial tree has positive degree. Replacing  $A_v$  by  $2^{h(v)}$  proves the claim. Thus occupied leaves pay the required slack, while unoccupied ambient leaves have unit weight and require no payment.  $\square$

**5.5. Consolidation to one ambient root.** We prove a pointwise statement about heights before applying the estimator. A set of vertices is called connected here if the unique tree path between any two of its vertices stays in the set. This agrees with connectedness of the induced subgraph, by uniqueness of paths in a tree.

**Lemma 5.8** (Source partition). *The vertices of  $T$  can be partitioned into connected sets  $P_i$  with vertices  $s_i \in P_i$  such that*

$$h(v) = d_T(s_i, v) \quad (v \in P_i).$$

*Proof.* For every  $v$  of positive height, choose once and for all an incoming auxiliary arrow  $p(v) \rightarrow v$  with  $h(p(v)) = h(v) - 1$ . Such an arrow exists by taking the last edge of a longest directed path ending at  $v$ : its prefix has length  $h(v) - 1$ , and (21) at the level of heights prevents the height of its penultimate vertex from being larger.

Iterating  $p$  decreases height by exactly one each time and ends at a vertex  $s(v)$  of height zero after  $h(v)$  steps. No vertex repeats, so this predecessor chain is the unique tree path from  $v$  to  $s(v)$  and  $h(v) = d_T(s(v), v)$ . Every vertex on the chain has the same terminal source, since its subsequent predecessor choices are fixed. Each nonempty fibre  $P_s = \{v : s(v) = s\}$  therefore contains  $s$  and the full path from each of its vertices to  $s$ . The path between two vertices of  $P_s$  is contained in the union of their paths to  $s$ , hence stays in  $P_s$ . The nonempty fibres give the required connected partition.  $\square$

**Lemma 5.9** (Merging height-dominated parts). *Let  $A, B$  be disjoint nonempty connected vertex sets joined by an edge  $xy$ , with  $x \in A$  and  $y \in B$ . Suppose  $a \in A$ ,  $b \in B$  and  $g : V(T) \rightarrow \mathbb{N}$  satisfy*

$$g(v) \leq d_T(a, v) \quad (v \in A), \quad g(v) \leq d_T(b, v) \quad (v \in B).$$

*Then  $A \cup B$  is connected, and one of  $a, b$  satisfies the same domination on all of  $A \cup B$ .*

*Proof.* Paths within the two sets, together with  $xy$ , connect their union. If  $z \in A$  and  $w \in B$ , concatenate the path from  $z$  to  $x$  in  $A$ , the edge  $xy$ , and the path from  $y$  to  $w$  in  $B$ . Disjointness makes this a simple path, so it is the unique tree path and

$$(25) \quad d_T(z, w) = d_T(z, x) + 1 + d_T(y, w).$$

Put  $\alpha = d_T(a, x)$  and  $\beta = d_T(b, y)$ . If  $\beta \leq \alpha + 1$ , then for every  $v \in B$  the triangle inequality and (25) give

$$g(v) \leq d_T(b, v) \leq \beta + d_T(y, v) \leq \alpha + 1 + d_T(y, v) = d_T(a, v).$$

The pre-existing domination on  $A$  is unchanged, so  $a$  works. Otherwise  $\beta > \alpha + 1$ , which implies  $\alpha \leq \beta + 1$ ; the same argument with the two sets exchanged shows that  $b$  works.  $\square$

**Proposition 5.10** (Height-dominating root). *There exists a vertex  $r \in V(T)$  such that*

$$(26) \quad h(v) \leq d_T(r, v) \quad (v \in V(T)).$$

*Proof.* Begin with the connected source partition of Lemma 5.8; each part is height-dominated by its own source. If there is more than one part, connectedness of  $T$  provides an edge joining two parts: for example, follow a path from a vertex in one part to a vertex outside it and take its first exit edge. Lemma 5.9, with  $g = h$ , merges those two parts and supplies a root that height-dominates their union. The resulting sets are still a partition into connected, height-dominated parts. Each merge reduces the number of parts by one, so finitely many merges leave the whole vertex set and one dominating root.  $\square$

**Theorem 5.11** (Arbitrary-defect upper bound). *For every configuration  $C$  on a finite tree  $T$  with at least two vertices,*

$$(S_v(C) \leq 0 \text{ for all } v) \implies |C| \leq \text{estim}(T) - 1.$$

*Proof.* For  $C = 0$ , the bound follows from the positive estimator formula. Otherwise the preceding construction applies. Choose the root  $r$  from Proposition 5.10. Since  $2^x$  increases with  $x$ , Lemma 5.7 gives

$$\begin{aligned} |C| &\leq L + \sum_{\deg_T(v) > 1} \deg_T(v) 2^{h(v)} \\ &\leq L + \sum_{\deg_T(v) > 1} \deg_T(v) 2^{d_T(r, v)} \\ &\leq L + \max_{s \in V(T)} \sum_{\deg_T(v) > 1} \deg_T(v) 2^{d_T(s, v)} = \text{estim}(T) - 1. \end{aligned}$$

The final equality is the estimator identity (4). Every quantity in this calculation uses the ambient tree.  $\square$

Together with the exact score criterion (Theorem 3.3), Theorem 5.11 implies that every non-stackable configuration has mass at most  $\text{estim}(T) - 1$ , and hence every configuration of mass  $\text{estim}(T)$  is stackable.

## 6. THE STACKING FORMULA

One final point is required because the stacking number uses exact sizes.

**Lemma 6.1.** *Let  $G$  be a connected graph on  $n \geq 2$  vertices. If  $t \geq 2$  and every configuration of mass  $t$  is stackable, then  $t > n$  and every configuration of mass  $t + 1$  is stackable.*

*Proof.* If  $2 \leq t \leq n$ , put one pebble on each of  $t$  distinct vertices. This configuration admits no legal move and is not stacked, a contradiction. Now let  $|C| = t + 1$ . Since  $t + 1 > n$ , some vertex has at least two pebbles. It has a neighbour by connectedness and  $n \geq 2$ . One legal move produces a configuration of mass  $t$ , which is stackable by hypothesis. Prepend that move to a stacking sequence.  $\square$

*Proof of Theorem 1.1.* Theorem 5.11 implies that every configuration of mass  $\text{estim}(T)$  is stackable. By (5), this is an admissible threshold at least two, so  $\text{stack}(T) \leq \text{estim}(T)$ . On the other hand, Corollary 4.2 provides a nonstackable configuration of mass  $\text{estim}(T) - 1$ . If a universal exact size  $t \geq 2$  were smaller than  $\text{estim}(T)$ , repeated application of Lemma 6.1 would make every configuration of mass  $\text{estim}(T) - 1$  stackable. This contradiction proves the reverse inequality.  $\square$

The proof also gives an exact extremal statement:

**Corollary 6.2.** *For every nontrivial finite tree,*

$$\max\{|C| : C \text{ is not stackable}\} = \max\{|C| : S_v(C) = 0 \text{ for every } v\} = \text{estim}(T) - 1.$$

*Proof.* Every zero-score configuration is nonstackable by Theorem 3.3. The upper bound is Theorem 5.11, and the zero-score obstruction attains it.  $\square$

For the singleton tree, every positive configuration is already stacked. Consequently  $\text{stack}(K_1) = 2$  under the source threshold convention, while (1)–(3) give  $\text{estim}(K_1) = 1$ .

## 7. FORMAL VERIFICATION AND REPRODUCIBILITY

The complete theorem, including the message characterization, obstruction, weighted cancellation and exact-size threshold argument, has been formalized in Lean. The headline declaration is

`TreeStack.stack_eq_estim_of_two_le_card.`

Its hypothesis and conclusion are, respectively,

$$2 \leq \text{Fintype.card } V, \quad \text{TreeStack.stack } T = \text{TreeStack.estim } T,$$

for a finite tree  $T$  on  $V$ . The formalization proves the result for arbitrary finite trees, without an order bound.

The source and reproduction instructions are available in the public repository [3]. The immutable formal-verification revision is

4d4969703a9f0ca7a51cbe7edf0f0338cc95cafb.

It pins Lean 4.35.0-rc2 and a specific Mathlib revision. The theorem-level audit records only the standard axioms `propext`, `Quot.sound` and `Classical.choice`. The proved declaration is separate from Palomar’s protected Challenge statement. Palomar mechanical verification succeeded, including comparison with that statement and independent NanoDa kernel replay. Its automated review identified no problems; registration has been requested and the public entry is pending. The verification-record link is reserved at `PALOMAR_VERIFICATION_URL_PLACEHOLDER`. This is a record of mechanical verification, not human peer review.

The repository includes a theorem-to-source correspondence for this paper, the pinned dependency files, and commands for the Lean build and axiom audit. It also provides independent Python implementations of legal reachability and branch messages, and bounded regression checks. Those computations support reproduction and testing; the proofs above do not depend on finite enumeration. Later publication commits preserve the immutable verification revision.

**Development provenance.** The work was developed with extensive AI assistance under the direction of the author, including mathematical exploration, Lean development, proof auditing and preparation of the exposition. The author selected the problem and scope and directed validation. This disclosure follows the development provenance recorded in the repository.

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